

4.9 Newton's Method Worksheet

1. Approximate $\sqrt{6}$ to four decimal places.

2. Approximate e^2 to four decimal places.

3. Approximate $\sqrt[n]{e}$ to five decimal places.

4. Find all solutions of the equation $\frac{1}{x} = 1 + x^3$ (up to six decimal places).

5. Consider the function $f(x) = x^3 - x - 3$.

a) Show that Newton's Method does not find a root if $x_1 = 0$. What does the sequence of approximation do here? Round decimals to six places.

b) Show that Newton's Method can find a root if $x_1 = 2$. Find the root accurate to six places.

c) What do parts (a) and (b) tell you about the selection of x_1 ?